

SPECIAL EVENTS

PHASES OF THE MOON

|      | FULL MOON   | NEW MOON    |
|------|-------------|-------------|
| 2020 | November 30 | November 15 |
| 2021 | November 19 | November 4  |
| 2022 | November 8  | November 29 |
| 2023 | November 27 | November 13 |
| 2024 | November 15 | November 1  |
| 2025 | November 5  | November 20 |
| 2026 | November 24 | November 9  |
| 2027 | November 14 | November 28 |

PLANETS

**2020: November 10** Mercury is at greatest morning elongation, magnitude  $-0.3$ .

**2023: November 3** Jupiter is at opposition, magnitude  $-2.9$ . At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

**2024: November 16** Mercury is at greatest evening elongation, magnitude  $-0.1$ .

**2026: November 20** Mercury is at greatest morning elongation, magnitude  $-0.3$ .

**2027: November 4** Mercury is at greatest morning elongation, magnitude  $-0.3$ .

ECLIPSES AND TRANSITS

**2021: November 19** A partial eclipse of the Moon is visible from North and South America, northern Europe, eastern Asia, Australia, and the Pacific.

**2022: November 8** A total eclipse of the Moon is visible from Asia, Australia, the Pacific, and North and South America.

# NOVEMBER

Cassiopeia lies overhead for northern observers, as the Milky Way runs from Cygnus in the west to Gemini in the east. The large figures of Pisces, the Fish, and Cetus, the Sea Monster or Whale, are spread across the equatorial region of the sky, while in the southern sky, the Large and Small Magellanic Clouds are high up.

## NORTHERN LATITUDES

### THE STARS

All the main characters in the Perseus and Andromeda myth (see p.368) are on display in the November evening sky. Cetus contains a remarkable variable star, Mira (see p.285). It is easily visible to the naked eye when at maximum brightness, every 11 months or so, but the rest of the time it fades out of sight. High in the west is the Square of Pegasus, with the stars of the Summer Triangle lower in the northwest.

### DEEP-SKY OBJECTS

Two open star clusters, NGC 457 and NGC 663, are easy to see with binoculars in Cassiopeia. Even better are NGC 869 and 884, a pair known as

the Double Cluster, embedded in the Milky Way between Perseus and Cassiopeia. The Andromeda Galaxy, M31 (see pp.312-313), remains high up this month.

### METEOR SHOWERS

The Taurids have a broad peak in the first week of the month, when around 10 meteors an hour may be seen coming from the region south of the Pleiades Cluster. Although not numerous, the meteors are long-lasting and often bright. A second meteor shower in November, the Leonids, radiates from the head of Leo,

### THE PRINCESS, HERO, KING, AND QUEEN

Joined in Greek myth, Andromeda (right), Perseus (bottom), Cepheus (top), and Cassiopeia (center) appear together in northern skies in November.

reaching a peak around November 17. Usually, no more than 10 meteors per hour are seen, but surges of activity occur every 33 years or so. High activity is not expected again until around 2032.

3 A.M.

MIDNIGHT

6 A.M.

9 A.M.

